

Answer Key — The Invisible Living World: Beyond Our Naked Eye

Final answers with brief reasons.

Objective

- Q1** (b) **Robert Hooke** — Hooke named cells from cork in 1665.
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- Q2** (c) **chloroplast** — chloroplasts (and a cell wall) are found only in plant cells.
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- Q3** (a) **nucleoid** — bacteria have a nucleoid, with no nuclear membrane.
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- Q4** (b) **virus** — viruses are acellular and reproduce only inside a host.
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- Q5** (c) **Lactobacillus** — Lactobacillus ferments milk into curd.
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- Q6** (a) **True** — that is the correct order of organisation.
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- Q7** **microbiology** — for first seeing living microbes.
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- Q8** **nitrogen** — Rhizobium fixes atmospheric nitrogen.
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Short answer

- Q9** Plant cell has a cell wall and chloroplasts; animal cell lacks both (plant has one large vacuole, animal small/none).
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- Q10** They are acellular and multiply only inside a living host cell.
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- Q11** (i) Yeast → carbon dioxide; (ii) Lactobacillus → lactic acid.
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Long / application

- Q12** Soil bacteria and fungi decompose dead plant/animal matter into simple, nutrient-rich manure that returns nutrients to soil; Rhizobium in legume root nodules fixes nitrogen from air, so crop rotation with legumes naturally enriches the soil.
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- Q13** Hooke named the cell from cork and wrote Micrographia (1665); Leeuwenhoek first saw living microbes (Father of Microbiology). Light microscope $\approx 100\text{--}400\times$; electron microscope $\approx 10,00,000\times$.
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